Stanley Nelavala

Backend and Platform Software Engineer
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Open to relocation: SF Bay Area, New York, Seattle

Summary

Software engineer who owns systems end to end. At a 10-person hospital operations startup I designed the service-to-service API that every platform module uses, built the hospital design service and the iOS and Android asset app from scratch, and cut dev and test server runtime from 168 to about 53 hours a week. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate. I also build AI-agent and reliability projects with public code and tests.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; all numbers from dev and test environments) *Huntington, IN*

- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Built RxDesign, the hospital design service, backend and web (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine that places rooms on a 0.5 ft grid with doors, exits and flow lines; 220+ offline tests on the API.
- Built the RxAsset iOS and Android app (React Native, Expo) with scan-to-search and an on-device voice assistant (no cloud call); set up separate iOS and Android GitHub Actions release pipelines with automatic TestFlight submission.
- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Search pages never skip or repeat rows, tested with rows inserted, deleted and tied between pages.
- Cut dev and acceptance server and database (ECS, RDS) runtime to about 53 hours a week instead of 168 with automated weekday start and stop. Standardized malware scanning of file uploads (GuardDuty) across 9 app modules.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session. Led labs and office hours for Python, NLP and Parallel Programming.

Projects (public code, tests)

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis, Terraform* GitHub

- LangGraph case workflow with typed state, a Postgres checkpointer, human approval that resumes after restart, and step and token budgets; 62 deterministic replay evals, prompt-injection tests, PII redaction and an append-only audit log.
- Hybrid retrieval (pgvector and full-text search fused by RRF) with access labels; failover between two LLM provider adapters with a circuit breaker and token cost records.

Payments Ledger Service | *Java 25, Spring Boot, PostgreSQL, Flyway, Kafka API, Prometheus, Grafana, k6* GitHub

- Three user-facing SLIs with 28-day error budgets and burn-rate alerts tested with promtool; a scripted failure lab (Kafka down, Postgres restart, poison event, latency injection) with balance checks after each.
- Rolled forward and back by image digest across an expand-and-contract Flyway migration; k6 SLO thresholds as a gate.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, Go, Rust
Backend and data: FastAPI, Spring Boot, REST, gRPC, PostgreSQL, Redis, Kafka API, migrations (Alembic, Flyway)
AI and agents: LangGraph, MCP, hybrid retrieval and RAG, deterministic LLM evals, guardrails
Frontend, mobile, cloud: React, Next.js, React Native, Expo, AWS (ECS, RDS, S3, SQS), Terraform, Docker, GitHub Actions
Testing and reliability: pytest, Jest, Playwright, k6, OpenTelemetry, Prometheus, Grafana, SLOs

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)